INCH-POUND

MS53233A
W/AMENDMENT 1
1 August 2023
SUPERSEDING
MS53233A
31 August 2007

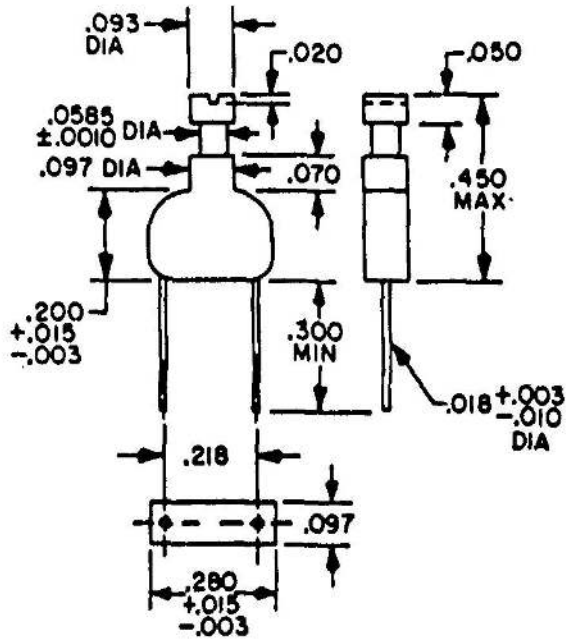
MILITARY SPECIFICATION SHEET

COILS, RADIO FREQUENCY, ENCAPSULATED, VARIABLE, MICRO-MINIATURE, IRON CORE, TYPES LT11V077 TO LT11V097, INCL

Inactive for new design.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification and MIL-PRF-15305.



INCHES	mm	INCHES	mm
.001	.03	.070	1.78
.003	.08	.093	2.36
.010	.25	.097	2.46
.015	.38	.200	5.08
.018	.46	.218	5.54
.020	.51	.280	7.11
.050	1.27	.300	7.62
.0585	1.49	.450	11.43

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm).

FIGURE 1. Dimensions and configuration.

AMSC N/A
FSC 5950

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REQUIREMENTS:

Design, construction, and physical dimensions: See figure 1.

Weight: .25 gram, maximum.

Operating temperature range: -55° to +105°C.

Ambient temperature: +90°C maximum.

Temperature rise: 15°C.

Power dissipation: 225 mW maximum.

Lead material: -1 through -12 is .018 (0.46 mm) diameter copper.
-13 through -21 is .018 (0.46 mm) diameter gold-plated Kovar.

Terminal pull: 2 pounds, minimum.

Tuning torque: .005 - .2 in. oz.

Stop torque: .2 in. oz. max.

Altitude: 70,000 feet.

Shock, specified pulse: MIL-STD-202-213, test condition I, is applicable.

Dielectric withstanding voltage:

At sea level: MIL-STD-202-301, test voltage 1,000 V rms minimum.

At reduced barometric pressure: MIL-STD-202-105, test condition C, test voltage 200 V rms minimum.

Resistance to soldering heat: MIL-STD-202-210, test condition B.

Electrical characteristics: See tables I and II.

Inductance: See table I.

Q values: See table I.

Self-resonant frequency (SRF): See table I.

DC resistance (DCR): See table I.

Marking: Marking shall be as specified in MIL-PRF-15305 except that the marking shall be on the unit package or container.

Part or Identifying Number (PIN): MS53233 - (dash number from table I).

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TABLE 1. Electrical characteristics (initial).

Dash Number MS53233	Inductance L Max. µH	Inductance L Min. µH	Q Min. At L Max.	Test Frequency (MHz)	SRF Minimum (MHz)	DC resistance At 25°C, max (Ohms)	Rated DC current, maximum (mA)
-1	.025	.020	50	200	1,000	.02	250
-2	.051	.035	40	100	700	.03	250
-3	.076	.040	40	100	615	.10	250
-4	.10	.07	30	25	500	.3	250
-5	.15	.11	30	25	420	.35	200
-6	.22	.16	30	25	400	.4	180
-7	.27	.19	30	25	400	.5	180
-8	.33	.165	30	25	320	.6	140
-9	.47	.34	30	25	290	.65	100
-10	.56	.40	30	25	250	.75	100
-11	.68	.50	30	25	240	.85	100
-12	.82	.59	30	25	180	.9	100
-13	1.0	.50	25	7.9	140	.4	125
-14	1.5	1.0	25	7.9	135	.8	100
-15	2.2	1.2	25	7.9	100	1.4	80
-16	2.7	1.95	25	7.9	60	1.6	75
-17	3.3	2.2	25	7.9	60	1.8	65
-18	4.7	2.5	25	7.9	50	2.2	60
-19	5.8	2.9	25	7.9	38	2.8	58
-20	6.8	3.4	25	7.9	36	3.0	55
-21	8.2	4.6	25	7.9	25	3.0	50

TABLE II. Electrical characteristics (final).

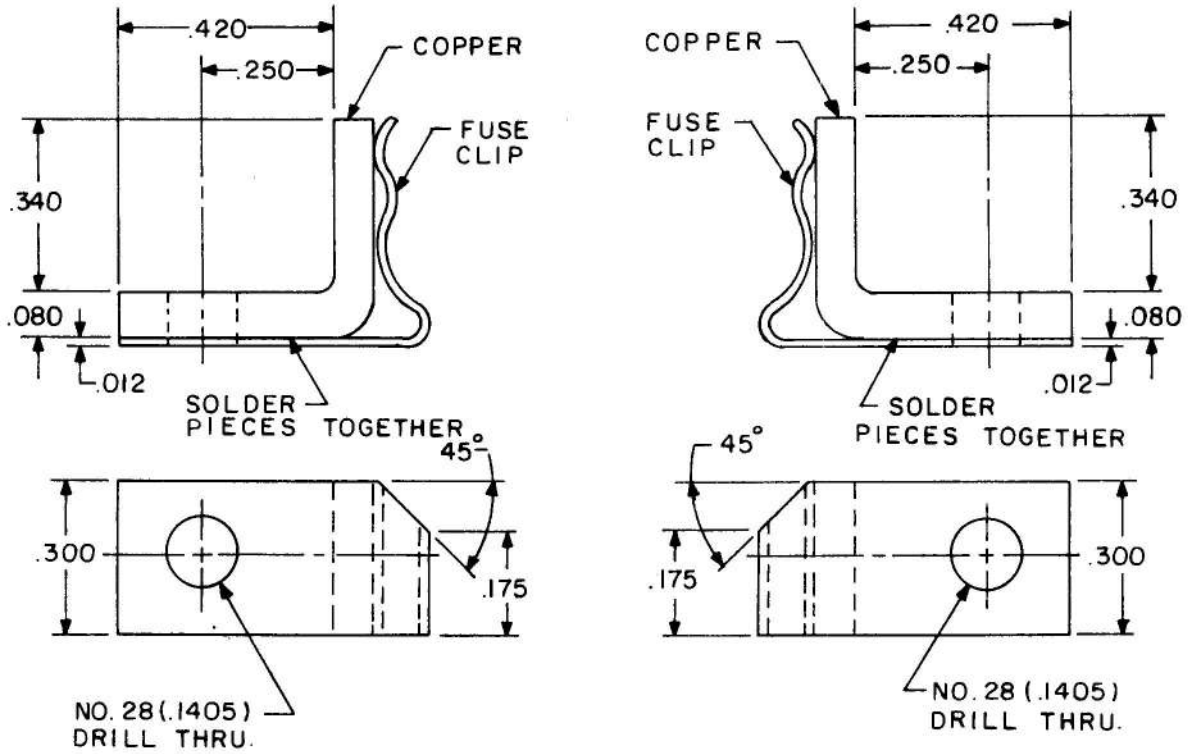
Inspection group	Allowable variation from Initial measurement	Allowable variation from Initial measurement	Allowable percent from specified minimum value in electrical characteristics (initial) table	Allowable percent from specified minimum value in electrical characteristics (initial) table
	Inductance (percent)	DC resistance	Self-resonant frequency	Q
Qualification inspection				
Group II	±5	---	---	-10
Group III	±5	±(2% +.001 ohm)	-5	-10
Group IV	±5	±(2% +.001 ohm)	-10	-10
Conformance inspection group C				
Subgroup I	±5	---	---	-10
Subgroup II	±5	±(2% +.001 ohm)	-10	-10
Subgroup III	±5	±(2% +.001 ohm)	-5	-10

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Application notes:

1. The test fixture in the diagram following shall be used for electrical measurements. Inductance values are effective inductance as indicated on a HP260A, HP190A or equivalent Q meter, when tested in the test fixture. Add 5 percent to Q reading to account for loss of Q in the test jig.
2. Polarization during moisture resistance test is not applicable.
3. Coils are held rigidly by the body during vibration and mechanical shock testing.
4. Resistance to soldering heat test, per MIL-STD-202-210, test condition B, is applicable.
5. For dielectric withstanding voltage, barometric pressure and insulation resistance units shall be placed on Flat metal plate with leads insulated from surface. Measurement of dielectric withstanding voltage, barometric pressure and insulation resistance shall be between the leads of the coil connected together and the metal plate.
6. Screw core assembly shall be set at maximum specified inductance value indicated in the electrical characteristics table (initial), prior to all inspection tests. This setting shall not be changed until electrical characteristics (final) measurements are performed.

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INCHES	MM
.012	0.30
.080	2.03
.1405	3.57
.175	4.45
.250	6.35
.300	7.62
.340	8.64
.420	10.67

FIGURE 2. Test fixture for electrical measurements.

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Referenced documents. In addition to MIL-PRF-15305, this document references the following:

MIL-STD-202-105 MIL-STD-202-210 MIL-STD-202-213 MIL-STD-202-301

Custodians:
Army – CR
Navy - EC
Air Force - 85
DLA – CC

Preparing activity:
DLA – CC

Project 5950-2023-020

Review activities:
Army – AR, MI
Navy – AS, MC, OS, SH
Air Force – 19

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